PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

GSD CarPass Depot App

In partial fulfillment for the requirements in Capstone 1 and Research Subject

GALVE, FRITZ MICHAEL G.

LADRILLO, CARL JUSTIN

PEREZ, GODWIN

SIMBA, VIA MARIE

TAMBA, MON WILBERT

System Development Track

Second Semester

2025-26

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

As the number of faculty members, administrative staff, and students who own and use vehicles continues to increase, educational institutions face growing challenges in managing parking facilities effectively. The rising volume of vehicles entering and exiting campus premises can lead to congestion, parking violations, and difficulties in monitoring vehicle movements. During peak hours, the manual checking of car passes at campus entry and exit points often causes traffic buildup, resulting in longer waiting times for vehicle owners and an increased workload for security personnel. These issues not only affect the convenience of campus users but also raise concerns regarding safety, security, and the efficient utilization of available parking spaces.

Furthermore, when parking-related issues arise, such as illegal parking or vehicles obstructing designated areas, guards and General Services Department (GSD) staff often face challenges in promptly contacting vehicle owners. The lack of an efficient communication and monitoring system can delay the resolution of parking violations and hinder the enforcement of campus parking regulations. As a result, the overall

management, safety, and accessibility of campus parking facilities may be negatively affected, emphasizing the need for a more efficient GSD CarPass Depot App.

At the University of Iloilo, the General Services Department (GSD) plays a crucial role in maintaining the efficiency, safety, and functionality of campus operations. The department is responsible for a wide range of services, including building maintenance, campus safety and security, management of lost and found items, and the implementation of vehicle entry and parking policies. Through these responsibilities, the GSD helps ensure that the university environment remains organized, secure, and conducive to learning and daily operations.

One of the GSD's key responsibilities is managing vehicle entries and parking allocations within the university. Currently, the university relies on a traditional manual car pass system, requiring applicants to physically visit multiple offices and complete paper-based procedures to obtain vehicle permits. While this process has been functional in the past, the growing number of vehicles entering the campus has highlighted several challenges, including longer processing times, increased administrative workload, and traffic congestion during peak hours due to manual sticker verification at the gates. In addition, the absence of a centralized digital repository for vehicle records makes it difficult to monitor real-time campus occupancy, verify authorized vehicles efficiently, and maintain accurate records for administrative purposes.

As institutions continue to adopt digital technologies to improve operational efficiency, reliance on physical stickers, paper-based records, and manual verification procedures has become increasingly ineffective in addressing the demands of modern campus management. These limitations can slow administrative processes and reduce the effectiveness of vehicle monitoring and parking enforcement.

Moreover, a well-managed vehicle access system is essential to the smooth daily operations of the university. The timely arrival and departure of faculty members, staff, and students are critical for maintaining academic schedules and administrative activities. Delays resulting from inefficient parking and vehicle verification procedures can negatively affect productivity and disrupt institutional operations. Therefore, adopting a technology-driven and centralized vehicle management system can significantly improve operational efficiency while providing a safer, more secure, and more convenient experience for all faculty members, staff, and students.

1.2 Statement of the Problem

The University of Iloilo's General Services Department (GSD) currently relies on a manual, paper-based GSD CarPass Depot App that has proven increasingly inadequate in addressing the demands of a growing campus population. The absence of a centralized and technology-driven solution has resulted in a range of operational inefficiencies that affect not only the administrative functions of the GSD but also the daily experiences of faculty

members, administrative staff, and students who own and use vehicles within the campus. As the volume of registered vehicles continues to rise, the limitations of the existing system have become more pronounced, manifesting in processing delays, security concerns, traffic congestion, and difficulties in enforcing campus parking regulations. The persistence of these challenges underscores the critical need for a modernized GSD CarPass Depot App that can effectively respond to the evolving needs of the university community. Specifically, this study seeks to address the following problems:

Inefficient Car Pass Application Process — The existing system requires applicants to physically visit multiple offices and complete paper-based procedures, resulting in longer processing times and a heavier administrative workload for GSD personnel.

Traffic Congestion at Entry and Exit Points — Manual sticker verification during peak hours causes significant traffic buildup at campus gates, leading to extended waiting times for vehicle owners and increased burden on security personnel.

Lack of a Centralized Digital Vehicle Repository — The absence of a unified digital system makes it difficult to monitor real-time campus parking occupancy, efficiently verify authorized vehicles, and maintain accurate and up-to-date vehicle records for administrative use.

Inadequate Enforcement of Campus Parking Regulations — Without a technology-driven monitoring system, enforcing parking policies and addressing violations in a timely and systematic manner remains a persistent challenge for the GSD.

Disruption to Academic and Administrative Operations — Delays caused by inefficient vehicle verification and parking management negatively affect the punctuality and productivity of faculty, staff, and students, ultimately disrupting the smooth daily operations of the university.

1.3 Objectives of the Study

1.3.1 General Objective

This study aims to design, develop, and implement a centralized, technology-driven and Parking Management System for the University of Iloilo's General Services Department (GSD) that will comprehensively address the growing inefficiencies and operational challenges associated with the current manual, paper-based approach to vehicle access control and parking administration. Recognizing the critical role that an efficient vehicle management system plays in sustaining the smooth and uninterrupted flow of daily academic and administrative operations, the study seeks to harness the capabilities of modern digital technology to transform and modernize the way the university handles vehicle entry, parking allocation, permit processing, and regulation enforcement within its campus premises. By transitioning to a fully digital system, the study intends to streamline the car pass application and approval process, establish a centralized digital repository for all vehicle-related records, and introduce a faster and more reliable vehicle verification mechanism that minimizes traffic congestion at campus entry and exit points during peak hours, thereby reducing the workload of security personnel and improving the overall flow of vehicle traffic within and around the university.

1.3.2 Specific Objectives

Specifically, the study aims to:

- **Develop a QR Code-Based Digital Car Pass Generation System** — Design and implement a digital car pass system that allows GSD staff to generate, manage, and issue QR code-based vehicle permits containing key vehicle information such as the plate number, owner name, owner email, and validity period, effectively replacing the existing manual, paper-based sticker system.

- **Implement an Automated QR Code Distribution via Email** — Enable the system to automatically deliver generated QR codes directly to vehicle owners through email, eliminating the need for applicants to physically visit multiple offices to claim their vehicle permits and reducing the administrative workload of GSD personnel.

- **Design a QR Code Scanning and Verification Mechanism for Security Guards** — Develop a scanning interface that allows security guards to quickly verify the validity of vehicle QR codes at campus entry and exit points, automatically detecting the status of each QR code as accepted, expired, or revoked, and logging each scan in real time to minimize manual checking and reduce traffic congestion during peak hours.

- **Establish a Centralized Digital Repository for Vehicle Records and Entry Logs** — Create a unified digital database that stores all QR code records, vehicle information, and entry and exit history, enabling authorized personnel to efficiently retrieve, monitor,

and manage vehicle data and campus parking occupancy at any given time.

- **Provide an Administrative Dashboard for Real-Time Parking Monitoring and Reporting** — Develop an admin dashboard that displays real-time data on parking slot availability, total vehicle entries and exits, active QR codes, and recent activity logs, enabling the GSD administration to make informed decisions regarding parking management and space utilization.

- **Enable Role-Based Access Control for System Users** — Implement a secure, role-based access control mechanism that assigns distinct permissions and functionalities to three user roles — Admin, Staff, and Guard — ensuring that each user can only access the features and data relevant to their designated responsibilities within the system.

1.4 Scope and Delimitation

Scope

This study covers the development and implementation of the GSD CarPass Depot App, a web-based application designed exclusively for the General Services Department (GSD) of the University of Iloilo. The system aims to modernize and streamline the department's vehicle access control and parking management operations through the use of QR code-based vehicle permits and digital record management.

The system accommodates three user roles with distinct functionalities. The Admin manages system user accounts, including the creation, viewing, and deletion of Admin, Staff, and Guard accounts; configures and adjusts total parking slot capacity; monitors real-time parking occupancy; views vehicle entry and exit activity logs; and generates reports through a comprehensive administrative dashboard. The Staff generates and issues QR code-based vehicle permits, manages vehicle owner information such as plate numbers and email addresses, delivers QR codes to vehicle owners via automated email, renews and reactivates expired QR code permits, and views the complete history of all generated QR codes. The Guard scans and verifies QR codes at campus entry and exit points using a dedicated scanning interface, logs each scan transaction with its corresponding status (accepted, expired, or revoked), and views personal scan history.

The GSD CarPass Depot App covers the generation and issuance of QR code-based vehicle permits containing essential information, including the vehicle plate number, owner name, owner email address, and permit validity period. An automated email delivery mechanism transmits the approved QR code-based vehicle permit directly to the registered email address of the vehicle owner upon generation by GSD staff. The system also supports the renewal and reactivation of expired QR code permits, allowing GSD staff to extend the validity of existing vehicle passes without generating entirely new QR codes.

The GSD CarPass Depot App provides real-time monitoring of parking slot availability and occupancy, automatically updating parking status based on QR code scan data recorded at campus entry and exit points. A centralized digital repository maintains all vehicle records, QR code permits, ownership information, and vehicle entry and exit history, accessible only to authorized GSD personnel. Activity logs and reports documenting vehicle entry and exit

transactions, QR code scan results, and overall parking utilization are generated by the system and are accessible to the Admin for administrative and reporting purposes. Profile management is also supported, allowing system users to update their account email address and password credentials.

The GSD CarPass Depot App is built using the Flask Python web framework for the backend, with an HTML, CSS, and JavaScript frontend, and a MySQL relational database for data storage and management. It does not have a dedicated mobile application; however, the QR code scanning interface used by security guards may be accessed through a mobile web browser, as the scanning feature utilizes the device's camera for QR code reading. The system requires a stable internet connection for all core functionalities, including QR code generation, email delivery, scanning and verification, and real-time parking monitoring. A secure authentication mechanism restricts access to authorized users only, with role-based permissions ensuring that each user can access only the features relevant to their designated role. The GSD CarPass Depot App is designed exclusively for deployment and use within the operational context of the University of Iloilo's General Services Department and is not intended for use by vehicle owners or the general public.

Delimitation

The study is bounded by the following limitations in terms of the scope and coverage of the **GSD CarPass Depot App**.

The system does not cover the online submission of car pass applications by vehicle owners. Applicants are still required to manually accomplish application forms, pay the corresponding car pass fees, and personally submit their documentary requirements to the GSD office prior to QR code permit generation. Accordingly, the system does not include an

automated payment processing feature or online collection of car pass fees, as the collection, verification, and recording of fees remain a manual responsibility of the GSD and are outside the scope of the system.

The system does not incorporate physical detection of vehicles using sensors, cameras, or any hardware-based vehicle detection technology for monitoring actual parking space utilization within the campus premises. Parking occupancy tracking is based solely on QR code scan data recorded at campus entry and exit points. As such, the system does not automatically detect, flag, or report parking violations, unauthorized parking, or vehicles occupying restricted areas. Physical monitoring and enforcement by security personnel remain necessary for these concerns. Real-time GPS tracking or location monitoring of vehicles within the campus premises is likewise not supported.

The system does not include a notification or alert system directed to vehicle owners regarding the status of their QR code permits, such as upcoming expiration reminders, renewal notices, or parking violation alerts. The system also does not support integration with third-party systems such as the university's enrollment system, human resource system, or any other existing institutional database. Furthermore, the system does not cover the generation of printed physical car pass stickers or any other form of physical vehicle permit, nor does it cover the management of pedestrian access, visitor passes, or non-motorized vehicles such as bicycles within the campus premises.

The system is accessible only via a web browser and does not include a dedicated mobile application for any of the three user roles on Android or iOS platforms. All core functionalities of the system, including QR code generation, email delivery, scanning and verification, and real-time parking monitoring, require a stable internet connection, as offline or

local network operation is not supported. Finally, the **GSD CarPass Depot App** is designed exclusively for the operational requirements of the University of Iloilo's General Services Department and is not intended for deployment in other institutions without significant modifications to accommodate different parking management structures and policies.

1.5 Significance of the Study

Significance of the Study

This study holds considerable significance as it addresses the real and pressing operational challenges faced by the University of Iloilo's General Services Department in managing vehicle access and parking within the campus. The development and implementation of the **GSD CarPass Depot App** is expected to bring meaningful improvements to the operational efficiency, security, and convenience of campus vehicle management. The following are the primary beneficiaries of this study:

General Services Department (GSD). The GSD stands as the primary beneficiary of this study. The system significantly reduces the administrative burden associated with the manual processing, issuance, and distribution of vehicle permits by digitizing the generation and delivery of QR code-based car passes. With a centralized digital repository for vehicle records and a real-time dashboard for monitoring parking occupancy and activity logs, the GSD will be better equipped to manage campus vehicle access efficiently, enforce parking policies, and maintain accurate and organized records without relying on paper-based procedures.

GSD Staff. The system directly improves the day-to-day workflow of GSD staff members by providing a streamlined digital interface for generating, issuing, renewing, and managing QR

code-based vehicle permits. The automated email delivery of approved car passes eliminates the need for staff to manually distribute physical permits, allowing them to dedicate more time and attention to other essential administrative responsibilities within the department.

Security Guards. Security personnel stationed at campus entry and exit points will benefit from a faster and more reliable vehicle verification mechanism. The QR code scanning interface enables guards to instantly determine the validity status of a vehicle permit as accepted, expired, or revoked, minimizing the time spent on manual sticker verification and reducing the occurrence of traffic congestion during peak hours. This also lessens the physical workload of guards and enhances the overall security of campus entry and exit operations.

Faculty Members and Administrative Staff. Faculty and administrative staff who own and use vehicles within the campus will experience a more convenient and efficient car pass issuance process. The automated delivery of QR code permits directly to their registered email addresses eliminates the need to make multiple visits to the GSD solely to claim their vehicle permits, saving time and effort while ensuring timely receipt of their approved passes. Additionally, the QR code renewal feature allows vehicle owners with expired permits to have their passes reactivated more efficiently without undergoing the full application process again.

University Administration. The University of Iloilo's administration will benefit from the improved organizational efficiency and enhanced security that the system brings to campus operations. The availability of real-time data on parking occupancy, vehicle entries and exits, and permit activity provides administrators with valuable insights for informed decision-making regarding parking infrastructure, policy enforcement, and resource allocation. The system also supports the university's broader institutional goal of adopting modern digital technologies to

improve operational efficiency, enhance campus safety and security, and provide a better environment for all members of the university community.

Future Researchers and Developers. This study serves as a foundational reference and practical resource for future researchers and developers who may seek to further enhance, expand, or build upon the GSD CarPass Depot App. The system's modular architecture, role-based access control design, and documented codebase provide a solid and scalable foundation for future development efforts. Future developers may extend the system by incorporating additional features such as online car pass application and submission, automated payment processing, hardware-based vehicle detection, mobile application development, or integration with other existing university systems. The insights, methodologies, and lessons documented in this study can likewise guide future researchers in addressing similar challenges in other institutional or organizational contexts, contributing to the broader body of knowledge in the field of campus management systems and information technology development.

1.6 Definition of Terms

Definition of Terms

The following terms are defined both conceptually and operationally to provide clarity and a common understanding of the key concepts used throughout this study:

Vehicle Access and Parking Management System. *Conceptual* — A vehicle access and parking management system refers to an integrated set of tools, technologies, and procedures designed to regulate, monitor, and control the entry, exit, and parking of vehicles within a defined

area or facility, with the goal of improving security, efficiency, and space utilization (Rodrigue, 2020). *Operational* — In this study, the Vehicle Access and Parking Management System refers to the web-based application developed for the General Services Department of the University of Iloilo that manages the generation and issuance of QR code-based vehicle permits, monitors real-time parking occupancy, and records vehicle entry and exit transactions within the campus premises.

QR Code (Quick Response Code). *Conceptual* — A QR code is a two-dimensional matrix barcode that stores information in a pattern of black and white squares and can be scanned using a digital device such as a smartphone or barcode reader to quickly retrieve encoded data (Denso Wave, 2020). *Operational* — In this study, QR code refers to the unique, digitally generated code embedded in each vehicle permit issued by the GSD staff through the system, which contains the vehicle owner's information and permit validity details and is scanned by security guards at campus entry and exit points for verification purposes.

Car Pass. *Conceptual* — A car pass is an official authorization document or permit issued by a governing authority or institution that grants a vehicle owner the right to access, enter, and park within a designated area or facility (Cambridge Dictionary, 2023). *Operational* — In this study, a car pass refers to the QR code-based digital vehicle permit generated and issued by GSD staff through the system to authorized faculty members, administrative staff, and students of the University of Iloilo, granting them the right to enter and park within the campus premises.

General Services Department (GSD). *Conceptual* — A general services department is an organizational unit within an institution responsible for managing and overseeing a wide range of support and administrative services, including facilities maintenance, security, and the implementation of institutional policies and procedures (Merriam-Webster, 2022). *Operational* —

In this study, the General Services Department (GSD) refers to the specific department of the University of Iloilo responsible for managing vehicle access, parking allocation, and the issuance of car pass permits within the campus, and which serves as the primary end-user and beneficiary of the developed system.

Real-Time Monitoring. *Conceptual* — Real-time monitoring refers to the continuous and immediate observation, tracking, and recording of data or activities as they occur, enabling users to access up-to-date information and respond promptly to changing conditions (Techopedia, 2021). *Operational* — In this study, real-time monitoring refers to the system's capability to instantly update and display the current parking occupancy, vehicle entry and exit counts, and QR code scan activity on the administrative dashboard as transactions are recorded at campus entry and exit points.

Role-Based Access Control (RBAC). *Conceptual* — Role-based access control is a security mechanism that restricts system access and functionalities to users based on their assigned roles within an organization, ensuring that individuals can only perform actions and view information relevant to their designated responsibilities (Ferraiolo & Kuhn, 1992). *Operational* — In this study, role-based access control refers to the system's security feature that assigns distinct permissions and functionalities to three user roles — Admin, Staff, and Guard — preventing unauthorized access to features and data outside the scope of each user's designated role within the GSD.

Web-Based Application. *Conceptual* — A web-based application is a software program that is hosted on a remote server and accessed by users through a web browser over the internet or a local network, without requiring installation on the user's device (Laudon & Laudon, 2020). *Operational* — In this study, web-based application refers to the GSD CarPass Depot App

developed for the General Services Department of the University of Iloilo, which is accessible through a web browser by authorized Admin, Staff, and Guard users to perform their respective vehicle management functions.

Parking Occupancy. *Conceptual* — Parking occupancy refers to the proportion or number of parking spaces within a facility that are currently in use at a given point in time, often used as a measure of parking demand and space utilization (Shoup, 2011). *Operational* — In this study, parking occupancy refers to the number of vehicles currently present within the campus parking area as tracked and displayed by the system's administrative dashboard, based on the QR code scan data recorded at campus entry and exit points.

Automated Email Delivery. *Conceptual* — Automated email delivery refers to the process by which a software system automatically generates and sends electronic mail messages to designated recipients upon the occurrence of a specific trigger or event, without requiring manual intervention from the sender (Chaffey, 2019). *Operational* — In this study, automated email delivery refers to the system's feature that automatically transmits the approved QR code-based vehicle permit to the registered email address of the vehicle owner immediately upon its generation by GSD staff, eliminating the need for manual distribution of physical car passes.

QR Code Renewal. *Conceptual* — QR code renewal refers to the process of extending or updating the validity period of an existing QR code, allowing it to remain functional and authorized beyond its original expiration date (ISO/IEC 18004, 2015). *Operational* — In this study, QR code renewal refers to the system feature that enables GSD staff to update the expiry date of an existing vehicle permit QR code, reactivating expired car passes and allowing vehicle

owners to continue using their previously issued QR codes without the need for a completely new permit to be generated.

Entry and Exit Logs. *Conceptual* — Entry and exit logs are systematic records that document the movement of individuals or vehicles into and out of a facility, typically including details such as the time, date, and identity of the person or vehicle involved (Black, 2019). *Operational* — In this study, entry and exit logs refer to the digital records automatically generated and stored by the system each time a vehicle's QR code is scanned at a campus gate by a security guard, capturing the scan timestamp, QR code details, vehicle plate number, and whether the transaction was an entry or exit.

Centralized Digital Repository. *Conceptual* — A centralized digital repository is a single, unified storage system that consolidates and organizes digital data and records from multiple sources, making them accessible to authorized users from one central location (Arms, 2000). *Operational* — In this study, centralized digital repository refers to the system's unified database that stores and maintains all vehicle permit records, QR code information, vehicle owner details, and entry and exit transaction logs, which can be accessed and managed by authorized GSD personnel through the web-based system.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Omkar et al. (2025) proposed a Smart QR Vehicle Access System with User-Defined Time Selection and ID Verification, designed to address persistent weaknesses in traditional vehicle access control methods such as manual verification, RFID cards, and biometric systems. The researchers noted that RFID-based systems remain vulnerable to cloning, while biometric authentication often introduces processing delays and raises privacy concerns among users. To overcome these limitations, the proposed system allows a registered user to scan a QR code displayed on the vehicle's screen, select a desired usage time, and submit an access request to the vehicle owner for approval. The owner then receives a real-time notification and may either accept or deny the request, and if the approved time period expires, the user is required to select a new duration and resubmit the request, ensuring that access remains tightly bounded to an explicitly authorized window. The system architecture relies on a touchscreen display, a QR scanner, and a microcontroller unit, supported by a cloud-based backend (Firebase) for real-time data handling, with Python and OpenCV used for backend processing and QR code scanning, respectively. The researchers concluded that by integrating user-defined time selection with real-time owner approval, the system enhances both the security and flexibility of vehicle access management, reducing reliance on manual oversight while preventing

unauthorized use.

Luciano et al. (2023) developed PARADAJuan, a web-based parking lot management system designed to assist parking lot owners in efficiently monitoring vehicle entries and exits as well as tracking the income generated from their facilities. The study was motivated by the practical challenges parking lot owners face in manually overseeing space availability and revenue, problems that a digital monitoring system could streamline. To build the application, the researchers employed multi-paradigm programming languages, intentionally combining more than one programming style and its associated language constructs so that each component of the system could be developed using the approach best suited to it. Specifically, the system was built using HTML5 and CSS3 for the front-end structure and styling, JavaScript for interactivity, PHP for server-side logic, MySQL for database management, and Flutter for additional application development. In terms of methodology, the study utilized developmental research methods, an approach in which the product-development process itself is systematically analyzed, described, and evaluated, with the final product assessed for functionality and effectiveness. The researchers reported that the development process was successful, with the system functioning as intended in supporting the monitoring of parking space usage and associated revenue tracking, supporting the broader argument that effective software systems for vehicle and facility management often rely on the deliberate integration of multiple tools chosen for their suitability to specific subsystems.

Khazri et al. (2019) proposed a model and accompanying set of system diagrams to define a new reservation and queuing system for remote laboratories used in educational contexts. The study was grounded in the recognition that remote laboratories play an increasingly important role in education, particularly for institutions that cannot afford to purchase, maintain, or staff physical laboratory equipment and space. The researchers observed that existing user management systems for remote laboratories were limited in scope and frequently incompatible with the variety of remote lab interface development platforms in use across institutions. To address this gap, they developed a new user management framework that integrates both a reservation system and a priority-based queuing mechanism, modeled using Unified Modeling Language (UML) to maintain design consistency and ensure adaptability across different programming languages and technical environments. Within the proposed framework, user requests for laboratory access are prioritized based on factors such as availability and predefined session durations, allowing the system to allocate limited laboratory time across multiple competing users in a structured and equitable manner. The researchers argued that this kind of optimized scheduling can significantly reduce access waiting times while simultaneously improving the overall utilization rate of laboratory resources. This reservation-and-queuing logic parallels the time-selection and approval mechanisms found in QR-based vehicle access systems, suggesting that design principles from remote laboratory scheduling may offer useful insights for structuring similar access-request workflows in other domains, such as shared vehicle and facility access

Sravanthi and Rajender (2025) developed a QR Code-Based Intelligent Parking System with Feedback Analytics aimed at addressing the growing challenges of urban congestion and inefficient parking management within smart city environments. The system incorporates secure login functionality for both regular users and administrators, dynamic slot visualization for real-time availability information, and QR code-based authentication selected specifically to minimize manual intervention and reduce entry and payment delays. A central feature distinguishing this study from earlier QR-based parking solutions is its feedback analysis module, which applies sentiment-based categorization to user feedback, enabling the system to assess overall satisfaction and present these insights through graphical visualizations for administrators. The system was built using the Django web framework with MySQL as the underlying database, chosen to ensure secure data handling and reliable real-time performance. The researchers conducted simulation-based testing and reported that the system was effective in improving parking allocation efficiency, user satisfaction, and overall system usability, positioning the study as a contribution to intelligent transportation infrastructure that extends the conventional access-control function of QR codes by layering in a user-experience feedback mechanism largely absent from other QR-access studies.

Jain et al. (2019) examined the persistent challenge of parking management in densely populated metropolitan areas, where the rapid and continuous growth in vehicle ownership has made finding an available parking space an increasingly frustrating experience for drivers. The researchers situated their work within the broader landscape of existing parking management technologies, noting that

systems such as Parking Guidance Information Systems (PGIS) and Geographic Information Systems (GIS) already provide drivers with a near real-time view of parking availability and allow for advance reservation of a space. In response, the study proposed a Smart QR Technology-based parking management system implemented through an Android application, integrating QR code technology with mobile and web-based infrastructure, drawing on Internet of Things (IoT) principles, network protocols, and a Web Map Server to support real-time spatial and parking-status information. The researchers argued that this QR-based approach offers benefits at both the individual level, by reducing the time and effort spent searching for parking, and the systemic level, by collectively helping to alleviate traffic congestion and reduce fuel consumption associated with vehicles circling in search of available spaces. The study reinforces a recurring theme across the QR-based access literature: that QR codes are frequently proposed as a low-cost, scalable substitute for more expensive or hardware-intensive access and guidance technologies, whether in vehicle entry systems, campus parking, or citywide parking management.

Ramdani, Ridwansyah, Wasid, and Irawan (2025) developed a web-based parking management system integrated with QR code technology and real-time notification features to address the challenges associated with traditional parking management methods. The study focused on improving the efficiency, security, and reliability of parking operations by automating vehicle authentication, monitoring parking activities, and providing immediate communication between users and administrators. The researchers recognized that conventional parking systems often suffer from issues such as manual record-keeping, slow vehicle verification processes, inaccurate data management, and limited monitoring capabilities, which can lead to congestion,

security risks, and operational inefficiencies. To overcome these problems, they designed a web-based platform that centralized parking information and enabled administrators to manage parking records through an online system. The integration of QR code technology served as a secure and efficient means of vehicle identification, wherein registered users were assigned unique QR codes that could be scanned upon entry and exit from the parking facility. This automated authentication process reduced the need for manual inspection, minimized human error, and accelerated vehicle processing times. In addition, the system incorporated real-time notification functionality that provided immediate updates regarding parking activities, including vehicle entry, exit, and access authorization, allowing both users and administrators to receive timely information and respond quickly to any concerns. The researchers emphasized that real-time communication contributes significantly to operational transparency and strengthens the overall security of parking facilities. Furthermore, the web-based system facilitated centralized data storage and management, enabling administrators to access accurate parking records, monitor vehicle movements, generate reports, and analyze parking utilization more effectively. The automated recording of transactions created a reliable audit trail that improved accountability and supported the monitoring of authorized and unauthorized vehicle activities. The findings of the study revealed that the integration of QR code authentication, web-based management, and real-time notifications significantly enhanced the efficiency of parking operations by reducing processing delays, improving record accuracy, strengthening security measures, and providing greater convenience to users. The researchers concluded that the adoption of digital technologies in parking management can transform traditional parking facilities into more intelligent, secure, and user-friendly environments. This study is highly relevant to the present research because it demonstrates the effectiveness of combining QR code technology with web-based systems to automate parking operations, improve monitoring

capabilities, and enhance access control. The findings provide valuable support for the development of a smart parking management system that aims to deliver efficient vehicle authentication, accurate data management, real-time monitoring, and improved user experience through the application of modern information and communication technologies.

2.2 Local Studies

According to Dela Cruz and Santos (2022), application-based systems developed in local academic settings improve process automation and data accuracy. Their study on intelligent web applications highlights the importance of user-centered design and system supervision. Reyes (2021) also emphasized that integrating data-driven decision-making enhances system reliability and usability.

2.3 Related Literature

Related Literature

The increasing number of vehicles in urban areas, educational institutions, and commercial establishments has intensified the need for efficient parking management systems. Traditional parking methods often rely on manual monitoring and paper-based records, which can lead to congestion, security issues, inaccurate data recording, and

inefficient utilization of parking spaces. To address these challenges, researchers have explored the integration of information technology, automation, QR code authentication, web-based platforms, and intelligent monitoring systems. These innovations aim to improve vehicle access control, optimize parking space allocation, and enhance the overall user experience.

One of the most relevant studies related to the development of modern parking management systems was conducted by Ramdani, Ridwansyah, Wasid, and Irawan (2025), who developed a web-based parking management system integrated with QR code technology and real-time notification features. The study was designed to address common challenges in conventional parking operations, including manual vehicle verification, inefficient monitoring of parking activities, delays in recording vehicle movements, and difficulties in communicating parking information to users and administrators. As parking facilities continue to experience increasing demand due to the growing number of vehicles, the researchers recognized the need for a technology-driven solution that could automate parking processes and improve operational efficiency.

The proposed system utilized a web-based platform that allowed administrators to manage parking records, monitor vehicle activity, and access parking information through a centralized database. The integration of QR code technology served as the primary authentication mechanism for vehicle entry and exit. Each registered user was assigned a unique QR code that could be scanned at designated access points,

enabling the system to verify authorization quickly and accurately. This approach eliminated the need for manual inspection of vehicle credentials and significantly reduced the time required to process vehicle access requests. By automating the verification process, the system minimized human intervention and reduced the likelihood of errors associated with manual record-keeping.

A key feature of the study was the incorporation of real-time notifications. Unlike traditional parking systems that rely on manual updates, the proposed system was capable of instantly transmitting information regarding vehicle entry, exit, and parking status. These notifications provided administrators with immediate awareness of parking activities, allowing them to respond promptly to potential issues or security concerns. At the same time, users benefited from timely updates regarding their parking transactions and vehicle access status. The researchers emphasized that real-time communication contributes to greater transparency and improves the overall user experience by ensuring that relevant information is delivered without delay.

The study also highlighted the importance of centralized data management in modern parking systems. Through the web-based platform, all parking-related information was stored and managed electronically, making it easier to retrieve records, generate reports, and analyze parking trends. The availability of accurate and organized data supported more effective decision-making and enabled administrators to monitor parking utilization more efficiently. Furthermore, the digital storage of records reduced the risk of data loss and improved the reliability of parking information compared to traditional

paper-based systems.

Another significant contribution of the study was its emphasis on security and accountability. The use of QR codes provided a secure method of identifying authorized users and tracking vehicle movements within the parking facility. Each transaction was recorded automatically in the system, creating an audit trail that could be reviewed when necessary. This feature enhanced accountability by ensuring that all vehicle activities were properly documented and traceable. The researchers noted that such capabilities are particularly important in environments where vehicle security and controlled access are priorities, such as universities, government institutions, business establishments, and private organizations.

The findings of Ramdani et al. (2025) revealed that the integration of QR code authentication, web-based management, and real-time notifications significantly improved the effectiveness of parking operations. The system reduced processing time for vehicle access, improved the accuracy of parking records, enhanced monitoring capabilities, and increased user convenience. Moreover, the researchers concluded that the adoption of digital technologies can help organizations manage parking facilities more efficiently while providing a higher level of service to users.

The relevance of this study to the present research lies in its demonstration of how multiple technologies can be integrated to create a comprehensive parking management solution. Similar to the proposed study, Ramdani et al. emphasized the use of QR codes for secure vehicle authentication and the importance of web-based platforms for

managing parking information. The inclusion of real-time notification features further illustrates how technology can enhance communication, monitoring, and security within parking environments. Consequently, the study provides valuable insights and serves as a strong foundation for the development of a smart parking management system that seeks to improve efficiency, security, and user satisfaction through the implementation of modern digital technologies.

Similarly, Jain et al. (2019) proposed a Smart QR-Based Parking System designed to streamline parking operations through the use of QR code technology. Their study focused on addressing common parking problems such as long waiting times, difficulty in vehicle identification, and inefficient parking space utilization. The researchers developed a system that generated unique QR codes for registered users, enabling fast authentication and access control. By replacing traditional manual verification procedures with digital authentication, the system reduced congestion at parking entrances and exits. Additionally, the study highlighted the ability of QR codes to facilitate real-time vehicle tracking and improve record management. The researchers concluded that QR code-based systems offer a cost-effective and reliable solution for parking management, particularly in areas with high vehicle traffic. Their findings support the growing adoption of QR technology as a practical tool for enhancing security, reducing delays, and improving the overall efficiency of parking facilities.

While parking management primarily focuses on vehicle access and space allocation, effective reservation and scheduling mechanisms are equally important for optimizing

resource utilization. In this regard, the study conducted by Khazri et al. (2019) provides valuable insights. The researchers designed and implemented a reservation system integrated with a queuing mechanism for remote laboratories. Their objective was to ensure fair and efficient allocation of limited resources among users. The study found that automated reservation systems significantly reduced scheduling conflicts and improved user satisfaction by providing transparent access to available resources. Furthermore, the queuing mechanism enabled the system to manage demand effectively and minimize resource wastage. Although the research was conducted in the context of remote laboratory management, its principles are highly relevant to smart parking systems. Reservation features can allow users to secure parking spaces before arrival, reducing uncertainty and congestion. Likewise, queuing algorithms can help manage peak-hour traffic and ensure equitable access to limited parking resources. The findings of Khazri et al. demonstrate the potential benefits of integrating reservation and scheduling functionalities into modern parking management systems.

A significant contribution to the field of parking management in the Philippine context was made by Luciano et al. (2023) through the development of PARADAJuan, a web-based parking lot management system. The study aimed to address common parking management challenges by utilizing multi-paradigm programming languages to create a comprehensive digital platform. The system provided features such as real-time parking space monitoring, user account management, automated record keeping, and centralized data storage. Through extensive testing and evaluation, the researchers found that the web-based system improved the efficiency of parking operations by

providing accurate and timely information to both administrators and users. The study also highlighted the importance of digital transformation in parking management, emphasizing that web-based platforms facilitate accessibility, scalability, and effective data management. The ability to monitor parking spaces in real time allows administrators to make informed decisions and respond quickly to changing conditions. Furthermore, the researchers noted that automated record generation reduced administrative workload and minimized the likelihood of human error. The findings suggest that web-based technologies are essential components of modern parking systems and can significantly improve service quality and operational performance.

Advancements in QR code applications have further expanded the capabilities of access control systems. Omkar et al. (2025) introduced a Smart QR Vehicle Access System that incorporates user-defined time selection and identity verification. Unlike conventional QR code systems that simply grant access based on authentication, this system allows administrators and users to define specific access schedules. The integration of time-based controls ensures that vehicles can only enter designated areas during authorized periods, thereby enhancing security and resource management. Additionally, the identity verification component strengthens the reliability of the authentication process by confirming the legitimacy of users before granting access. The study demonstrated that combining QR technology with scheduling and verification mechanisms can create a more robust and flexible access control system. The researchers concluded that such systems are particularly beneficial in institutions, gated communities, and organizations where controlled vehicle access is necessary. Their

findings indicate that QR-based access systems can be enhanced through the integration of additional security and scheduling features, making them more adaptable to complex operational requirements.

Another recent study that contributes to the development of intelligent parking systems was conducted by Sravanthi and Rajender (2025). The researchers proposed a QR code-based intelligent parking system integrated with feedback analytics to support smart urban mobility initiatives. Their system not only facilitated vehicle authentication and parking space management through QR codes but also incorporated mechanisms for collecting and analyzing user feedback. The feedback analytics component enabled administrators to evaluate system performance, identify recurring issues, and implement data-driven improvements. The study found that incorporating user feedback into parking management systems enhances service quality and increases user satisfaction. Furthermore, the researchers emphasized that data analytics can provide valuable insights into parking patterns, occupancy trends, and user behavior, allowing administrators to optimize parking operations more effectively. The integration of analytics into parking systems represents a significant advancement in the field, as it enables continuous improvement and supports evidence-based decision-making.

Collectively, these studies demonstrate the growing importance of technology-driven solutions in addressing parking management challenges. The research conducted by Farizy and Andrianto (2022) and Jain et al. (2019) highlights the effectiveness of QR code technology in improving access control, reducing processing times, and enhancing

security. Meanwhile, the study of Khazri et al. (2019) underscores the value of reservation and scheduling mechanisms in optimizing resource utilization and reducing congestion. Luciano et al. (2023) emphasize the role of web-based technologies in providing real-time monitoring, efficient data management, and improved administrative processes. Furthermore, the works of Omkar et al. (2025) and Sravanthi and Rajender (2025) illustrate how QR code systems can be enhanced through identity verification, time-based access control, and feedback analytics.

The literature collectively supports the development of a comprehensive smart parking management system that integrates QR code authentication, web-based monitoring, reservation functionality, real-time data management, and analytical capabilities. Such a system has the potential to improve security, maximize parking space utilization, reduce vehicle congestion, and enhance user satisfaction. These findings provide a strong theoretical and practical foundation for the present study and demonstrate the relevance of implementing intelligent parking technologies in modern parking environments.

Navalta (2022) developed a Vehicle Monitoring System with QR code technology for the Don Mariano Marcos Memorial State University, North La Union Campus in Bacnotan, La Union, aiming to improve the efficiency and accuracy of monitoring vehicle entry and exit within the campus premises. The study addressed common issues in traditional vehicle monitoring processes, such as manual logging, data inconsistency, slow verification procedures, and limited security in tracking vehicle movements. To resolve these concerns, the researcher implemented a QR code-based system that assigns

unique digital identifiers to authorized vehicles, allowing for faster and more reliable authentication at campus entry and exit points. When scanned, the QR code enables the system to automatically record vehicle information, including time of entry and exit, thereby reducing reliance on manual encoding and minimizing human error in data collection. The system also improves the security of campus vehicle monitoring by ensuring that only registered and verified vehicles are granted access, enhancing control over campus entry points. Furthermore, the digital nature of the system allows for better organization and storage of vehicle records, making it easier for administrators to retrieve data, generate reports, and monitor traffic flow within the university. The study found that QR code integration significantly improved the speed and accuracy of vehicle monitoring while also strengthening security measures within the campus environment. It also demonstrated that automated monitoring systems reduce administrative workload and provide more reliable data compared to traditional paper-based methods. The researcher concluded that QR code-based vehicle monitoring systems are effective tools for enhancing campus security and improving the management of vehicle-related activities. This study is relevant to the present research because it highlights the practical application of QR code technology in a real institutional setting, demonstrating its effectiveness in automating vehicle tracking, improving data accuracy, and strengthening access control. The findings support the development of similar smart parking and vehicle management systems that aim to enhance efficiency, security, and monitoring capabilities through the use of modern digital technologies.

2.4 Synthesis of the Reviewed Literature

Synthesis of the Reviewed Literature

The reviewed literature highlights the growing significance of technology-driven solutions in addressing the challenges associated with traditional parking management systems. As the number of vehicles continues to increase in educational institutions, commercial establishments, and urban communities, conventional parking methods have become insufficient in ensuring efficient vehicle access, security, and space utilization. The studies reviewed consistently emphasize the need for automated, intelligent, and user-centered parking systems that leverage modern technologies such as QR codes, web-based platforms, reservation mechanisms, and data analytics. Collectively, these studies provide a strong foundation for the development of a smart parking management system capable of improving operational efficiency and enhancing user experience.

A common theme that emerged from the reviewed studies is the effectiveness of QR code technology as a secure and efficient method of vehicle authentication and access control. Farizy and Andrianto (2022), Jain et al. (2019), Omkar et al.

(2025), and Sravanthi and Rajender (2025) all demonstrated the successful application of QR codes in parking and vehicle management systems. Their findings revealed that QR codes can significantly reduce the time required for vehicle verification, eliminate manual record-keeping processes, and minimize human errors. By providing unique digital identifiers for authorized users, QR codes enable faster entry and exit procedures while improving overall parking security. The repeated use of QR technology across multiple studies indicates its reliability, affordability, and practicality for parking management applications. These findings support the integration of QR code authentication as a core component of the proposed system.

Another significant pattern identified in the literature is the importance of automation in improving parking operations. Traditional parking systems often require manual monitoring and administration, which can lead to delays, inaccuracies, and inefficiencies. The studies conducted by Farizy and Andrianto (2022) and Luciano et al. (2023) demonstrated that automated systems can streamline parking processes by reducing the need for human intervention. Automated verification, monitoring, and record management not only improve operational efficiency but also enhance the accuracy and reliability of parking-related data. Automation further contributes to the reduction of congestion at entry and exit points, allowing parking facilities to accommodate

more users effectively. The literature therefore suggests that incorporating automation into parking management systems can lead to substantial improvements in both administrative performance and user satisfaction.

The reviewed studies also underscore the value of web-based technologies in modern parking management. Luciano et al. (2023) highlighted how web-based platforms facilitate real-time monitoring, centralized data storage, and efficient information management. Through web technologies, administrators can access parking data remotely, monitor parking space occupancy, and generate reports with greater accuracy and convenience. Additionally, web-based systems provide users with easier access to parking-related information, enhancing transparency and improving the overall parking experience. The findings indicate that web-based platforms play a critical role in ensuring scalability, accessibility, and effective communication between system administrators and users. As a result, the integration of web-based functionalities is essential for the development of a comprehensive smart parking management solution.

Reservation and scheduling capabilities emerged as another important aspect of efficient resource management. Although Khazri et al. (2019) focused on remote laboratory reservations rather than parking systems, their findings offer valuable insights that can be applied to parking management. The study demonstrated

that reservation systems improve resource allocation by allowing users to secure access in advance, thereby reducing conflicts and maximizing utilization. The incorporation of scheduling and queuing mechanisms also contributes to better management of limited resources during periods of high demand. Applying these concepts to parking management can help reduce vehicle congestion, minimize waiting times, and improve the overall efficiency of parking operations. The literature therefore supports the inclusion of reservation features as a means of enhancing convenience and optimizing parking space utilization.

Security is another recurring theme throughout the reviewed literature. Several studies emphasized that modern parking systems must go beyond simple space allocation and address concerns related to unauthorized access and vehicle safety. Omkar et al. (2025) demonstrated that combining QR code authentication with identity verification and time-based access controls significantly strengthens security measures. By ensuring that only authorized individuals can access designated parking areas during specific periods, such systems reduce the risk of unauthorized entry and improve accountability. Similarly, the QR-based systems developed by Farizy and Andrianto (2022) and Jain et al. (2019) contributed to improved access control and monitoring capabilities. These findings indicate that security-enhancing features are essential elements of effective parking

management systems and should be incorporated into future implementations.

In addition to security and efficiency, the reviewed literature highlights the growing importance of data collection and analytics in intelligent parking systems. Sravanthi and Rajender (2025) introduced a parking system that integrates feedback analytics, enabling administrators to evaluate user experiences and identify areas for improvement. The study demonstrated that data-driven decision-making can contribute to continuous system enhancement and increased user satisfaction. Similarly, web-based management systems provide opportunities to collect and analyze parking data, including occupancy rates, peak usage periods, and user behavior patterns. Such information can support strategic planning, resource allocation, and service optimization. The literature suggests that integrating analytical tools into parking management systems can transform them from simple operational platforms into intelligent decision-support systems.

The reviewed studies collectively indicate that effective parking management requires the integration of multiple technologies rather than reliance on a single solution. QR code authentication provides secure and efficient access control, while web-based platforms facilitate real-time monitoring and centralized management. Reservation mechanisms optimize resource allocation, identity

verification enhances security, and analytics support continuous improvement. Each study contributes a unique perspective and addresses specific aspects of parking management, but together they reveal a common goal: the creation of intelligent, efficient, secure, and user-friendly parking systems.

Despite the advancements presented in the literature, several gaps remain that warrant further investigation. Many of the reviewed studies focused primarily on individual system components, such as QR code authentication, reservation mechanisms, or web-based management, rather than integrating these features into a unified platform. Additionally, some studies were conducted in specific contexts, such as university campuses or controlled facilities, which may limit the generalizability of their findings. There is therefore a need for research that combines multiple technologies into a comprehensive parking management system capable of addressing a broader range of operational challenges. Furthermore, limited attention has been given to the integration of real-time monitoring, reservation features, security mechanisms, and user feedback analytics within a single system.

Overall, the reviewed literature establishes a strong theoretical and practical foundation for the present study. The findings consistently demonstrate that the integration of QR code technology, web-based platforms, automated monitoring,

reservation systems, access control mechanisms, and analytical tools can significantly improve parking management operations. These technologies contribute to enhanced security, reduced congestion, efficient utilization of parking spaces, accurate record management, and improved user satisfaction. Based on the synthesis of the reviewed studies, it can be concluded that a smart parking management system incorporating these features has the potential to address the limitations of traditional parking methods and provide a more effective solution for modern parking environments. Consequently, the present study seeks to build upon these findings by developing and evaluating a smart parking management system that integrates these proven technologies into a comprehensive and efficient platform.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study will use a **developmental research design**, focusing on the design, development,

and evaluation of the proposed application.

3.2 System Development Methodology

The **Agile Development Model** will be used, consisting of the following phases:

1. Requirements Analysis
2. System Design
3. Development
4. Testing
5. Deployment (Prototype)

3.3 System Architecture

The system will follow a client-server architecture. Users will interact through a web or mobile interface, while the server will handle data processing, machine learning predictions, and database management.

3.4 Tools and Technologies Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python / PHP / Node.js
- **Machine Learning:** Python (Scikit-learn)

- **Database:** MySQL
- **Development Tools:** Visual Studio Code

3.5 Data Gathering Techniques

- Surveys and questionnaires
- Interviews with potential users
- Review of existing dating platforms

3.6 Testing Procedures

- Unit Testing
- System Testing
- User Acceptance Testing

3.7 Evaluation Criteria

The system will be evaluated based on the following criteria:

- Functionality
- Usability
- Accuracy of Machine Learning Predictions
- Security
- Efficiency

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

User → Dating Application → Database

- User submits profile and preferences
- System processes data and generates matches
- Results are stored and retrieved from the database

Entity Relationship Diagram (ERD)

Entities:

- User (UserID, Name, Age, Preferences)
- Profile (ProfileID, UserID, Interests, BehaviorScore) ● Match (MatchID, UserID, MatchUserID, CompatibilityScore) ● Admin (AdminID, Name, Role)

Relationships:

- User creates Profile
- User generates Match

- Admin supervises User

Use Case Diagram

Actors:

- User
- Administrator

Use Cases:

- Register Account
- Create/Edit Profile
- View Matches
- Send Messages
- Monitor User Behavior (Admin)
- Manage Accounts (Admin)

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.

References

Arms, W. Y. (2000). *Digital libraries*. MIT Press. https://www.cs.cornell.edu/wya/DigLib/MS1999/index.html

Black, H. C. (2019). *Black's law dictionary* (11th ed.). Thomson Reuters.

Cambridge University Press. (2023). Car pass. In *Cambridge English dictionary*. https://dictionary.cambridge.org/dictionary/english/car-park

Chaffey, D. (2019). *Digital marketing: Strategy, implementation and practice* (7th ed.). Pearson Education. https://www.pearson.com/en-us/subject-catalog/p/digital-marketing/P200000004951

Denso Wave Incorporated. (2020). *What is a QR code?* https://www.qrcode.com/en/about/

Ferraiolo, D. F., & Kuhn, D. R. (1992). Role-based access controls. *Proceedings of the 15th National Computer Security Conference*, 554–563. https://csrc.nist.gov/CSRC/media/Publications/conference-paper/1992/10/13/role-based-access-controls/documents/ferraiolo-kuhn-92.pdf

International Organization for Standardization. (2015). *ISO/IEC 18004:2015 — Information technology — Automatic identification and data capture techniques — QR code bar code symbology specification*. https://www.iso.org/standard/62021.html

Laudon, K. C., & Laudon, J. P. (2020). *Management information systems: Managing the digital firm* (16th ed.). Pearson Education. https://www.pearson.com/en-us/subject-catalog/p/management-information-systems-managing-the-digital-firm/P200000005896

Merriam-Webster. (2022). General services. In *Merriam-Webster dictionary*. https://www.merriam-webster.com/dictionary/general%20services

Rodrigue, J. P. (2020). *The geography of transport systems* (5th ed.). Routledge. https://doi.org/10.4324/9780429346323

Shoup, D. (2011). *The high cost of free parking* (Updated ed.). Planners Press, American Planning Association. https://doi.org/10.4324/9781315174198

Techopedia. (2021). *Real-time monitoring*. https://www.techopedia.com/definition/31740/real-time-monitoring

Jain, S., Mishra, S. K., Bihani, M., & Goyal, P. (2019). Smart QR based parking system. *Journal of Computer Technology & Applications, 10*(3), 12–18.

Khazri, Y., Fahli, A., Moussetad, M., & Naddami, A. (2019). Design and implementation of a reservation system and a new queuing for remote labs. *International Journal of Online and Biomedical Engineering, 15*(12), 57–68. https://doi.org/10.3991/ijoe.v15i12.11098

Luciano, R. G., Cunanan, A. I., Mariano, R. P., Tavares, E. N. A., & Jacinto, M. R. L. (2023). PARADAJuan: A web-based parking lot management system designed and developed using multi-paradigm programming languages. *International Journal of Information Engineering and Electronic Business, 15*(3), 57–67. https://doi.org/10.5815/ijieeb.2023.03.05

Omkar, P. V., Supriya, G. H. V., Manikiran, N. V., & Manohar, K. (2025). Smart QR vehicle access system with user-defined time selection and ID verification. *International Journal of Innovative Research in Technology, 11*(10), 4475–4480.

Sravanthi, S., & Rajender, S. (2025). A QR code-based intelligent parking system with feedback analytics for smart urban mobility. *Fringe Multi-Engineering Proceedings, 1*(3), 26–38.

N. Ramdani, R. Ridwansyah, A. Wasid, and D. Irawan, “Web-Based Parking Management System with QR Code and Real-Time Notifications,” *IKRA-ITH Informatika: Jurnal Komputer dan Informatika*, 2025.

J. L. Navalta, “Vehicle Monitoring System with QR Code for the Don Mariano Marcos Memorial State University, North La Union Campus, Bacnotan, La Union,” University of the Philippines Open University, 2022.